

LECTURE 11: STACKELBERG COMPETITION

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WEEK 4

OUR LAST MODEL OF COMPETITION

- We are going to add a little spice to the Cournot (choosing quantity model)
- We will let one firm go first (just like our sequential games!)
- We call this our Stackelberg model of competition

WHY DOES SOMEONE GOING FIRST CHANGE THINGS?

- Often the first firm into the market keeps a high amount of market share. Consumers are reluctant to switch brands.
- iPhone was first into the true touch-screen phones and has held enormous share since then
- We are going to model this as a strategic phenomenon, entering first means that your rivals have to respond to your actions



WHY DOES SOMEONE GOING FIRST CHANGE THINGS?

Simultaneous games:

- Both players react to each other at the same time
- Nash Equilibrium when both firms' best response functions coincide/intersect with each other

Sequential games:

- Only the second mover responds to the first mover
- The first mover dictates how the game will play out
- Nash Equilibrium when the second mover does its best response to the first mover

WHY DOES SOMEONE GOING FIRST CHANGE THINGS?

How does this map into our models of firm competition?

Simultaneous games: (Cournot and Bertrand games)

- We find both players' best response functions
- Cournot/Bertrand Nash equilibria are at the intersection of the best response functions

Our new sequential Stackleberg game:

- We find the best response function of the second mover
- First mover considers the second mover's best response in their decisions

LET'S GO BACK TO OUR AIRLINE EXAMPLE

- American Airlines and United are competing to sell flights from Sacramento to LA
- Inverse demand for this route is $P = 339 - Q_D$
- Both airlines have no fixed costs and constant marginal costs of \$147
- But now let's say that **United** enters the market first, then **American** follows



HOW DO WE APPROACH THIS?

- 1 - Find market demand as a function of how much each firm supplies
- 2 - Find the best response function for our second mover - how much should the second mover sell as a function of other the first mover's quantities
- 3 - Maximise the first mover's profit, using the second mover's best response function

SOLVING THE AIRLINE EXAMPLE AS A STACKELBERG

These steps are like solving our subgames from bottom to top



Outcome: (United profit, American profit)

SOLVING THE AIRLINE EXAMPLE AS A STACKELBERG

1 - Market demand. We have market demand as a function of both airlines as:

$$P = 339 - q_U - q_A$$

SOLVING THE AIRLINE EXAMPLE AS A STACKELBERG

2 - Find American's (second mover) best response function:

American's profit function is:

$$\begin{aligned}\pi_A &= (339 - (q_U + q_A))q_A - 147 \times q_A \\ &= 339q_A - q_U q_A - q_A^2 - 147q_A\end{aligned}$$

American's best response function is:

$$\begin{aligned}\frac{d\pi_A}{dq_A} &= 339 - q_U - 2q_A - 147 = 0 \\ -2q_A &= -192 + q_U \\ q_A &= BR_A = 96 - \frac{1}{2}q_U\end{aligned}$$

SOLVING THE AIRLINE EXAMPLE AS A STACKELBERG

3 - Optimise United (our first mover's) profit using American's best response function:

- Where we see American's quantity, q_A , we substitute in American's best response function

$$\begin{aligned}\pi_U &= 339q_U - q_U \times q_A - q_U^2 - 147 \times q_U \\ &= 339q_U - q_U \times (96 - \frac{1}{2}q_U) - q_U^2 - 147 \times q_U \\ &= 339q_U - 96q_U + \frac{1}{2}q_U^2 - q_U^2 - 147 \times q_U \\ &= 339q_U - 96q_U + \frac{1}{2}q_U^2 - q_U^2 - 147 \times q_U \\ &= 96q_U - \frac{1}{2}q_U^2\end{aligned}$$

Intuition here: United knows how American will respond to its entry, so it maximises its profit already knowing that response!

And we optimise United's profit:

$$\begin{aligned}\frac{d\pi_U}{dq_U} &= 96 - q_U = 0 \\ q_U &= 96\end{aligned}$$

SOLVING THE AIRLINE EXAMPLE AS A STACKELBERG

- We have United, our first mover will sell 96 flights
- American, our second mover, will sell $q_A = 96 - \frac{1}{2} \times 96 = 48$ flights
- Compare this against our original Cournot game, where each would sell 64 each!
- United entering the market first lets them capture more the original market before American can enter!

WHY IS THE STACKELBERG EQUILIBRIUM DIFFERENT?

- Why does the first mover get to sell more than in the original Cournot (simultaneously choosing quantities) game
- The first mover gets to force the hand of the second mover - by moving first, the first mover can take more of the demand
- The second mover has to just take whatever's left
- In the original game, they're competing on a level playing field - both responding to each other simultaneously

